

BNKE: Kernel-Sensitive Shot Allocation for Projected Quantum Kernels Under Finite Measurement Budgets

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Abstract—Projected quantum kernels (PQK) are a promising approach to quantum machine learning, but their practical use is limited by finite measurement budgets on near-term hardware. Under restricted shots, different observables contribute unequally to kernel quality, yet measurements are typically allocated uniformly. We propose BNKE (Budget-aware Noisy Kernel Estimation), a weighted shot allocation framework redistributing a fixed budget by observable importance, derived from classical uncertainty, measurement variance, kernel sensitivity, or their combination. We derive a Neyman-type optimum for a weighted kernel estimation surrogate and show it never underperforms uniform allocation under this objective. Across five MedMNIST datasets, kernel sensitivity is the strongest allocation guide: the sensitivity-only variant beats uniform on all five over 30 paired runs (Holm-adjusted $p \leq 5.8 \times 10^{-4}$, $d_z = 0.70$ – 2.96), while variance-only and boundary-only weighting yield no consistent gain, and a separate 10-run-per-split analysis gives a positive mean paired difference on all 25 dataset-split combinations. Gains are largest at intermediate budgets and vanish at high per-entry shot counts. Because the weights use exact features, these results characterise an oracle-informed upper bound rather than a deployable procedure, and motivate estimating sensitivity from a measurement pilot.

Index Terms—Projected Quantum Kernels (PQK), shot allocation, finite measurement budgets, quantum machine learning, kernel sensitivity, surrogate loss, NISQ devices.

I. INTRODUCTION

Quantum kernel methods. Quantum kernel methods are a principled approach to applying near-term quantum hardware to supervised learning [1], [2]. A parameterized quantum circuit maps classical inputs to quantum states, whose inner products define a kernel evaluated by a classical support vector machine. This separation preserves the theoretical properties of kernel methods while enabling feature representations that may be difficult to reproduce classically [3].

Projected quantum kernels (PQKs). Projected quantum kernels (PQKs) provide a practical variant of this framework [4]. Instead of estimating full quantum state overlaps, PQKs construct feature vectors from single-qubit Pauli expectations $\{\langle X_q \rangle, \langle Y_q \rangle, \langle Z_q \rangle\}$ measured independently per qubit, followed by a classical similarity function.

The finite-shot problem. Expectation values are estimated from a finite number of measurements (shots). From S measurement shots, we get an unbiased estimate of each Pauli expectation f . The variance of this estimate is $(1 - f^2)/S$.

Since the budget B is fixed, it must be distributed across all $N \times M$ sample–observable pairs. In realistic settings (e.g., $N = 5,000$, $M = 18$, $B = 500,000$), uniform allocation leaves only a small number of shots per entry, so estimation noise becomes a primary source of error in the resulting kernel.

The allocation gap. Despite this constraint, PQK implementations typically rely on uniform measurement allocation. Framed in machine-learning terms, this is a measurement-budget allocation problem: given a fixed sampling budget for estimating a noisy kernel matrix, how should samples be distributed across entries so as to best preserve downstream classification? This is closely related to optimal experimental design and active learning, here applied to quantum feature estimation. This ignores heterogeneity across observables: some have higher variance, while others exhibit greater *kernel sensitivity*, i.e., small perturbations induce larger changes in the kernel matrix. Allocating shots accordingly is natural but has not been systematically studied for projected kernels. The main **contributions** of this work are as follows:

- **BNKE framework.** We present **BNKE** (Budget-aware Noisy Kernel Estimation), a weighted allocation framework for distributing measurement shots within a predetermined budget. BNKE accepts an importance score for each sample–observable pair and converts it into a shot allocation via a Neyman-type rule derived from a weighted surrogate objective. We consider importance scores derived from classical boundary uncertainty, measurement variance, kernel sensitivity, and their combination.
- **Theoretical characterization.** Under a weighted surrogate loss, we show the proposed allocation never underperforms uniform allocation. A first-order approximation motivates a composite sensitivity–variance weight, but the ablation is what shows sensitivity alone to be the dominant signal; the theory does not predict this on its own.
- **Empirical evaluation.** We test five allocation strategies on five MedMNIST datasets [5], 30 paired runs each. Sensitivity-only allocation beats uniform on every dataset (Holm-adjusted $p \leq 5.8 \times 10^{-4}$, $d_z = 0.70$ – 2.96); variance-only and boundary-based weighting do not. The BNKE-S weight is computed from exact features, with no labels, classifier outputs, or margins, so it should be read

as an oracle upper bound rather than a result achievable under hardware constraints.

- **Numerical stability analysis.** We find that PQK kernels are often ill-conditioned in practice, which affects the stability of SVM training. In these cases, ridge regularization plays an important role; without it, exact kernels can appear worse than noisy ones due to numerical effects.

Scope. Our setting is shot allocation for projected kernels under a fixed budget, simulated with synthetic shot noise. Everything runs through one pipeline: 6 qubits, a depth-1 circuit, an RBF kernel at $\gamma = 0.5$, and MedMNIST inputs cut to 6 PCA dimensions. We do not examine deeper circuits, larger systems, or alternative kernels, and we claim no quantum advantage; what we offer is measurement efficiency inside a pipeline held fixed throughout. More importantly, the sensitivity weights behind our strongest variant come from exact state-vector features, so the reported numbers are an oracle upper bound, not something a device could reproduce today.

A. Quantum kernels and feature-space formulations

Quantum kernel methods use similarities between quantum states as a kernel for a classical learner [1], [2], separating quantum feature construction from classical optimization.

B. Limits of quantum kernels

Prior work suggests that hard-to-simulate quantum feature maps are not enough to establish a practical advantage. Huang et al. [4] found that classical models can perform similarly when enough training data are available. Kübler et al. [3] further showed that obtaining informative quantum kernels may require a large number of measurements. Shaydulin and Wild [6] also found that kernel results depend strongly on bandwidth choice, making measurement cost an important part of the overall evaluation.

C. Hybrid PQC models and measurement efficiency

Parameterized quantum circuits are also used as hybrid learning models [7]–[9]; Jerbi et al. [10] compare such models, kernel methods, and data re-uploading on expressivity and resource use, orthogonal to our focus on measurement allocation. Most quantum-kernel studies either assume exact kernel values or use the same number of shots for every measurement, although finite-shot noise is unavoidable on NISQ hardware. Adaptive shot allocation has been used to reduce measurement cost in variational algorithms [11], but those methods mainly focus on estimating individual expectation values. Here, the problem is different: a single measurement budget must be distributed across many projected-kernel entries. To our knowledge, existing work has not allocated this budget according to how much each observable affects kernel estimation and the final classification result. This work is intended to address that gap.

II. PROBLEM FORMULATION

A. Projected Quantum Kernel Setup

Feature vector. Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ be a labeled dataset with $x_i \in \mathbb{R}^D$ and $y_i \in \{1, \dots, C\}$. Inputs are reduced

to n dimensions via PCA and rescaled to $[0, \pi]$. A depth-1 parameterized quantum circuit $U(x)$ encodes each input into an n -qubit state $|\psi(x)\rangle$. The projected quantum kernel (PQK) feature vector is formed from single-qubit Pauli expectations:

$$\mathbf{f}_i = [\langle X_q \rangle_i, \langle Y_q \rangle_i, \langle Z_q \rangle_i]_{q=1}^n \in \mathbb{R}^M, \quad M = 3n. \quad (1)$$

A classical RBF kernel is applied:

$$K_{ij} = \exp(-\gamma \|\mathbf{f}_i - \mathbf{f}_j\|^2), \quad \gamma = 0.5. \quad (2)$$

B. Finite-Shot Estimation

Finite-shot noise. Each expectation value $f_{ik} = \langle P_{q,k} \rangle_i$ is estimated from S_{ik} independent measurements. The estimator $\hat{f}_{ik}$ is the empirical mean of ± 1 outcomes, with variance:

$$\text{Var}(\hat{f}_{ik}) = \frac{1 - f_{ik}^2}{S_{ik}}. \quad (3)$$

Replacing exact features with $\hat{f}_{ik}$ perturbs the kernel entries in Eq. (2), which in turn affects the downstream SVM solution.

C. The Allocation Problem

Budget constraint. A fixed shot budget B must be distributed across all $N \times M$ sample–observable pairs:

$$\sum_{i=1}^N \sum_{k=1}^M S_{ik} = B, \quad S_{ik} \geq 1. \quad (4)$$

Note that the one-shot floor makes the problem feasible only for $B \geq NM$; budgets below this value are excluded from all experiments reported here. Here B is the training-side allocation budget; test-set measurement cost is fixed across methods and accounted for separately in Section V.

Uniform allocation sets $S_{ik} = \lfloor B/(NM) \rfloor$. At practical scales (e.g., $N = 5,000$, $M = 18$, $B = 500,000$), this yields only a few shots per entry, placing the system in a shot-limited regime. The objective is to allocate shots to reduce a *weighted surrogate* for kernel estimation error (Section IV), rather than classification accuracy directly. Empirically, improvements in this surrogate correlate with better downstream performance, but no direct optimization guarantee for accuracy is assumed.

III. BNKE: BUDGET-AWARE NOISY KERNEL ESTIMATION

A. Overview

BNKE is a weighted allocation framework: it accepts an importance score for each sample–observable pair and converts that score into a shot allocation. The score may be derived from classical boundary uncertainty, measurement variance, kernel sensitivity, or a combination of these; the variants studied below differ only in this choice. Fig. 1 illustrates the pipeline.

Importance weighting. BNKE assigns an importance weight w_{ik} to each sample–observable pair (i, k) and allocates shots using:

$$S_{ik}^* = B \cdot \frac{\sqrt{w_{ik}}}{\sum_{j,\ell} \sqrt{w_{j\ell}}}. \quad (5)$$

We study four non-uniform choices of w_{ik} , together with uniform allocation, and identify sensitivity-only weighting (BNKE-S) as the most effective in practice. The continuous

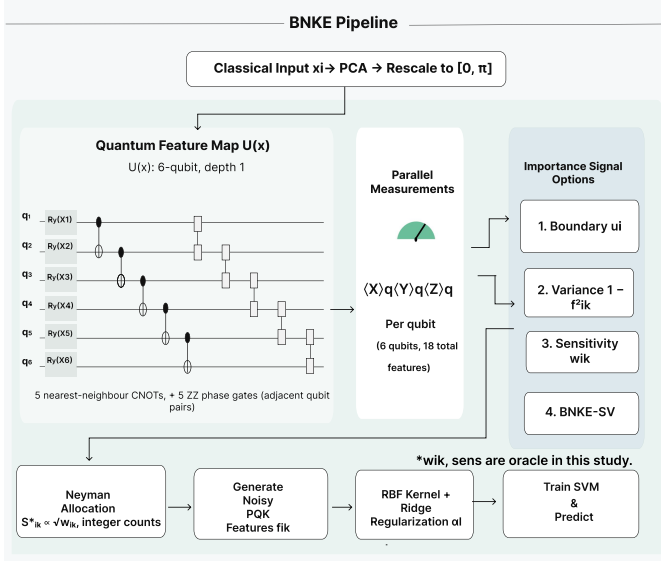


Fig. 1. BNKE pipeline. Six-qubit POK features feed one of four importance signals—margin, variance, sensitivity, or the sensitivity-variance product (BNKE-SV)—which a Neyman-type rule converts into the training-shot allocation. Signals are used singly, never chained. Variance and sensitivity are computed from exact features here, so they are oracle quantities; BNKE-S is the best finite-shot method.

solution is converted to integer shot counts using the largest-remainder method: each S_{ik}^* is floored to $\max(1, \lfloor S_{ik}^* \rfloor)$, so every entry gets at least one shot, and the leftover budget $\Delta = B - \sum_{i,k} \lfloor S_{ik}^* \rfloor$ is handed out one shot at a time to whichever entries have the largest fractional remainders until nothing is left. Entries with $w_{ik} = 0$ get only the one-shot minimum, and the budget saved from them feeds back into this redistribution. Uniform allocation works differently: every entry gets the same floored value $\lfloor B/(NM) \rfloor$, and whatever remains after that is simply left unused. We do not redistribute it there, because uniform allocation has no weight indicating which entry would benefit most from an extra shot, a choice the adaptive methods can make and uniform cannot.

B. Candidate Signal: Classical Boundary Uncertainty

Classical margin signal. Sample uncertainty is estimated with a linear SVM fitted to the exact POK feature matrix. This uses the same oracle feature access as the sensitivity and variance variants and consumes no shots from the allocation budget, but it is not a measurement-free classical pilot: the features themselves require measurement on hardware. Writing $m_i \geq 0$ for the margin of sample i (the absolute decision value in the binary case, the gap between the two largest decision values otherwise), the boundary weight is

$$u_i = \frac{1}{1 + m_i}, \quad (6)$$

normalized to unit mean, so that samples near the decision boundary receive higher weight. This signal is used only by BNKE-B, which consumes labels through the fitted margin classifier; BNKE-S does not use it and is label-free at allocation.

C. Candidate Signal: Kernel Sensitivity

Sensitivity measure. A first-order approximation of kernel perturbation yields:

$$w_{ik}^{\text{sens}} = \sum_{j=1}^N K_{ij}^2 (f_{ik} - f_{jk})^2. \quad (7)$$

Higher values indicate greater influence on the kernel matrix.

Equation (7) depends only on the feature matrix and the kernel: it uses no labels, no classifier, and no margin information. In this work it is computed from exact POK features in simulation, i.e. as an oracle quantity. In practice a pilot measurement pass would be required to approximate it, so the results reported here should be read as an upper bound on what sensitivity-guided allocation can achieve.

D. Allocation Variants

We study five allocation strategies under the same total budget B and identical downstream pipeline.

- **BNKE-U** (Uniform): $S_{ik} = \lfloor B/(NM) \rfloor$.
- **BNKE-V** (Variance-only): $w_{ik} = 1 - f_{ik}^2$, $S_{ik}^* \propto \sqrt{1 - f_{ik}^2}$.
- **BNKE-B** (Boundary-only): $w_{ik} = u_i$, $S_{ik}^* \propto \sqrt{u_i}$.
- **BNKE-S** (Sensitivity-only): $w_{ik} = w_{ik}^{\text{sens}}$, $S_{ik}^* \propto \sqrt{w_{ik}^{\text{sens}}}$.
- **BNKE-SV** (Sensitivity $\times$ Variance): $w_{ik} = w_{ik}^{\text{sens}}(1 - f_{ik}^2)$, $S_{ik}^* \propto \sqrt{w_{ik}^{\text{sens}}(1 - f_{ik}^2)}$.

The ablation evaluates the contribution of sensitivity, variance, and their combination. Results in Section V show that sensitivity is the dominant signal: BNKE-S and BNKE-SV both improve substantially over uniform allocation and are statistically indistinguishable from one another, while BNKE-V and BNKE-B provide no consistent improvement.

E. Ridge Regularization for Kernel Stability

POK kernels are often ill-conditioned, which affects numerical stability. We apply ridge regularization:

$$K_{\text{train}} \leftarrow K_{\text{train}} + \alpha I, \quad (8)$$

where α is selected by cross-validation and kept fixed across allocation methods (Section V).

IV. THEORETICAL ANALYSIS

We derive the optimal allocation rule under a weighted surrogate loss, compare it with uniform allocation, and use a first-order approximation to motivate kernel sensitivity as an importance weight. The analysis covers shot noise only and does not guarantee classification accuracy.

A. Measurement Statistics

Lemma 1 (Pauli Measurement Variance). *Let $P \in \{X, Y, Z\}$ be a single-qubit Pauli observable and let $f = \langle P \rangle \in [-1, 1]$ be its true expectation value under a pure quantum state. The estimator $\hat{f}$ formed by averaging S independent single-shot measurement outcomes is unbiased, with*

$$\text{Var}(\hat{f}) = \frac{1 - f^2}{S}. \quad (9)$$

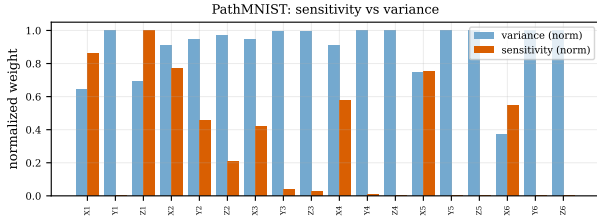


Fig. 2. Per-observable variance and kernel sensitivity on PathMNIST (both normalized). Sensitivity varies far more across observables than variance, and the two are not aligned.

Proof. Outcomes are i.i.d. in $\{-1, +1\}$ with mean f and second moment 1, so one shot has variance $1 - f^2$; averaging S gives the result. $\square$

Variance is largest at $f \approx 0$ and vanishes as $f \rightarrow \pm 1$, so highly polarized observables are cheaper to estimate—which explains the heterogeneous variance pattern in Fig. 2.

B. Optimal Allocation Under a Weighted Surrogate

Proposition 1 (Neyman Allocation is Optimal for the Surrogate). Define the weighted surrogate loss

$$L(\{S_{ik}\}) = \sum_{i=1}^N \sum_{k=1}^M \frac{w_{ik}}{S_{ik}}, \quad (10)$$

where $w_{ik} \geq 0$ are fixed importance weights (variance-only: $w_{ik} = 1 - f_{ik}^2$; sensitivity-only: $w_{ik} = w_{ik}^{\text{sens}}$; composite: $w_{ik} = w_{ik}^{\text{sens}}(1 - f_{ik}^2)$, Prop. 3). Subject to $\sum_{i,k} S_{ik} = B$ with $S_{ik} > 0$, the minimizer of L is

$$S_{ik}^* = B \cdot \frac{\sqrt{w_{ik}}}{\sum_{j,\ell} \sqrt{w_{j\ell}}}. \quad (11)$$

Proof. Stationarity of the Lagrangian $\sum_{i,k} w_{ik}/S_{ik} + \lambda(\sum_{i,k} S_{ik} - B)$ gives $S_{ik}^* \propto \sqrt{w_{ik}}$; normalizing to the budget yields Eq. (11). $\square$

Entries with $w_{ik} = 0$ receive the one-shot minimum, with the saved budget redistributed (Sec. III); the continuous solution is otherwise rounded to integer counts. This is the classical Neyman allocation [12] applied to quantum measurement scheduling, optimal with respect to L only.

Proposition 2 (Surrogate-Optimal Allocation Never Underperforms Uniform). Let L^* and L^{unif} be the surrogate losses under optimal and uniform allocation. Then $L^* \leq L^{\text{unif}}$, with equality iff all w_{ik} are equal.

Proof. Uniform allocation gives $L^{\text{unif}} = \frac{NM}{B} \sum_{i,k} w_{ik}$; optimal allocation gives $L^* = \frac{1}{B} (\sum_{i,k} \sqrt{w_{ik}})^2$. By Cauchy–Schwarz with sequences $\sqrt{w_{ik}}$ and 1,

$$\frac{L^*}{L^{\text{unif}}} = \frac{(\sum_{i,k} \sqrt{w_{ik}})^2}{NM \sum_{i,k} w_{ik}} \leq 1, \quad (12)$$

with equality iff all w_{ik} are equal. $\square$

The bound holds only under the weight set generating the allocation: it guarantees $L(w) \leq L^{\text{unif}}(w)$, not a cross-weight-set comparison, and bounds the surrogate rather than

accuracy, which we assess in Sec. V. This matters in Fig. 4, where all methods share one reference weighting, so a variant not optimizing it (e.g. BNKE-V) can exceed uniform there without contradicting Eq. (12). We verified numerically that the guarantee holds per method under its own weights.

C. Motivating Kernel Sensitivity as an Importance Weight

First-order approximation.

Proposition 3 (First-Order Sensitivity Approximation). For the RBF kernel $K_{ij} = \exp(-\gamma \|\mathbf{f}_i - \mathbf{f}_j\|^2)$, the coordinatewise contribution of perturbations in observable k at sample i , holding the remaining coordinates fixed at first order, is approximately

$$\mathbb{E}[\delta K_{ij}^2] \approx 4\gamma^2 K_{ij}^2 (f_{ik} - f_{jk})^2 \cdot \text{Var}(\hat{f}_{ik}), \quad (13)$$

where δK_{ij} is the perturbation to K_{ij} induced by shot noise in $\hat{f}_{ik}$ alone. Summing over j and substituting $\text{Var}(\hat{f}_{ik}) = (1 - f_{ik}^2)/S_{ik}$ motivates the composite importance weight

$$a_{ik} = w_{ik}^{\text{sens}} \cdot (1 - f_{ik}^2), \quad w_{ik}^{\text{sens}} = \sum_j K_{ij}^2 (f_{ik} - f_{jk})^2. \quad (14)$$

Using $w_{ik} = w_{ik}^{\text{sens}} \cdot (1 - f_{ik}^2)$ in Proposition 1 recovers the BNKE-SV allocation rule.

Proof. The partial derivative of K_{ij} with respect to f_{ik} is

$$\frac{\partial K_{ij}}{\partial f_{ik}} = -2\gamma K_{ij} (f_{ik} - f_{jk}).$$

A first-order Taylor expansion around exact features gives

$$\delta K_{ij} \approx -2\gamma K_{ij} (f_{ik} - f_{jk}) (\hat{f}_{ik} - f_{ik}).$$

Since $\hat{f}_{ik}$ is unbiased, taking expectations of the squared perturbation gives Eq. (13). Summing over j yields Eq. (14). $\square$

Scope of the approximation. Proposition 3 treats one coordinate at a time. In practice all NM coordinates carry noise simultaneously, but under independent, zero-mean estimation errors the cross terms between different coordinates cancel in expectation to first order, and only the single-coordinate terms of Eq. (13) survive. What matters is not how close f_{ik} and f_{jk} are, but whether the γ -scaled noise stays small enough for the local linear approximation to hold. A special case arises when $f_{ik} = f_{jk}$: the derivative in Eq. (7) vanishes, so the first-order weight is zero even though shot noise still perturbs that entry at second order. Near-equal feature values and low-shot regimes are therefore where the weight most understates a pair’s importance; Eq. (7) justifies the weight choice rather than guaranteeing kernel-MSE.

D. Empirical Validation of the Noise Model

Noise model validation. The empirical total feature MSE $\sum_{i,k} (\hat{f}_{ik} - f_{ik})^2$ closely tracks the variance-based prediction $\sum_{i,k} (1 - f_{ik}^2)/S$ in the moderate-shot regime, supporting the variance model. At low shot counts a single realization can deviate somewhat due to finite-sample fluctuation, and clipping near the physical bounds can further reduce observed MSE below the unclipped prediction.

Exact shot sampling vs. the clipped Gaussian. The clipped-Gaussian model of Eq. (15) is not algebraically identical to averaging S i.i.d. ± 1 outcomes, with the largest discrepancy in the 4–5-shot regime used in the main experiments. We therefore repeated the simulation with exact Bernoulli sampling, $n_+ \sim \text{Binomial}(S_{ik}, (1 + f_{ik})/2)$, $\hat{f}_{ik} = 2n_+/S_{ik} - 1$, comparing each model’s estimator error relative to the exact features rather than comparing single realizations directly. Over 20 repetitions at five shots per entry, RMSE is 0.408 (clipped Gaussian) vs. 0.423 (Bernoulli), a 4% difference. Method ordering is unchanged: on PathMNIST at $B = 500,000$ over five Bernoulli runs, mean accuracies are 0.273 (Uniform), 0.273 (BNKE-V), 0.318 (BNKE-S), 0.315 (BNKE-SV) (BNKE-B omitted; the check targets the sensitivity ranking). With five runs this is preliminary, not a formal test.

V. EVALUATION

A. Datasets

We evaluate on five MedMNIST v2 benchmarks [5]. Three are used as primary benchmarks: **PathMNIST** (9-class colorectal histopathology), **DermaMNIST** (7-class dermatoscopy), and **PneumoniaMNIST** (binary chest X-ray classification). Two further benchmarks, **OrganAMNIST** (11-class abdominal CT) and **BloodMNIST** (8-class blood cell microscopy), assess whether the effect extends beyond the primary three within the same pipeline. For each dataset, we sample up to $N_{\text{train}} = 5,000$ training examples and $N_{\text{test}} = 1,000$ test examples from the standard MedMNIST splits using a fixed random seed and sampling without replacement. PneumoniaMNIST contains only 4,708 training and 624 test images after the standard split, so the full available set is used. Sampling indices are fixed at seed 42 across all methods and runs, and all five datasets use an identical pipeline. Figure 2 illustrates the variance heterogeneity across PQK observables that motivates non-uniform shot allocation.

B. Preprocessing and Feature Extraction

Images are flattened and scaled to $[0, 1]$. PCA reduces the input to $n = 6$ dimensions, matching the number of qubits. Each principal component is then rescaled to $[0, \pi]$ using the training-set minimum and maximum, and the same scaling is applied to the test set.

PQK features are computed with an exact state-vector simulator implemented in PyTorch and GPU-accelerated, using a batch size of 500 state vectors. For each sample, the single-qubit Pauli expectations $\{\langle X_q \rangle, \langle Y_q \rangle, \langle Z_q \rangle\}_{q=1}^6$ are extracted, yielding $M = 18$ real-valued features.

C. Simulation Methodology

Near-term quantum devices are affected by gate errors, decoherence, and measurement noise, which currently limit reliable large-scale kernel computation [13], [14]. We therefore use exact state-vector simulation and inject shot noise analytically, following established practice in studies of finite-shot effects in quantum kernels and variational algorithms [4], [11], [15]. With $n = 6$ qubits each state vector has dimension $2^6 = 64$,

so the full 30-run ablation across five methods and five budget levels is inexpensive.

a) *Controlled shot-noise model.*: Exact Pauli expectations $f_{ik} = \langle P_{q,k} \rangle_i$ are perturbed according to

$$\hat{f}_{ik} = \text{clip}\left(f_{ik} + \mathcal{N}\left(0, \frac{1 - f_{ik}^2}{S_{ik}}\right), [-1, 1]\right). \quad (15)$$

This isolates the effect of finite-shot noise from other hardware-level confounds. Section IV-D validates this model against exact Bernoulli shot sampling.

b) *Circuit specification.*: The circuit $U(x)$ consists of three layers:

- 1) single-qubit R_y rotations with $\theta_q = x_q \in [0, \pi]$,
- 2) nearest-neighbour CNOT gates,
- 3) adjacent ZZ-type phase gates $e^{-ix_q x_{q+1} Z_q \otimes Z_{q+1}/2}$.

The circuit is intentionally shallow so that performance differences reflect shot allocation, not circuit expressivity.

D. Kernel Construction and Ridge Regularization

The PQK feature matrix is converted to an RBF kernel with $\gamma = 0.5$, and ridge regularization is applied to the training kernel as in Eq. (8). The parameter α is selected by 3-fold stratified cross-validation over

$$\alpha \in \{0, 10^{-4}, 10^{-3}, 0.01, 0.05, 0.1, 0.5, 1.0, 5.0, 10.0\}.$$

The selected values are $\alpha = 0.05$ for PathMNIST and PneumoniaMNIST, $\alpha = 0.01$ for DermaMNIST and OrganAMNIST, and $\alpha = 0.5$ for BloodMNIST. These values are then held fixed across all allocation methods on the same dataset.

E. Classifier and Budget Accounting

We use a kernel SVM with class-balanced sample weights and a maximum of 100,000 iterations. The regularization parameter C is selected by 3-fold stratified cross-validation, with a collapse-detection step penalizing values of C that predict fewer classes than appear in the validation fold. Under a total training budget B , each Pauli expectation is replaced by its noisy estimate per Eq. (15). Test features always receive uniform shots, $\lfloor B/((N_{\text{train}} + N_{\text{test}})M) \rfloor$ (72,000 at $B = 500,000$), so all methods face identical test-kernel noise. Throughout, B means the training budget B_{train} , the part redistributed by Eq. (11); since B_{test} is method-independent it takes no part in redistribution, and we report B_{train} as the only portion a method controls.

F. Measurement Budget and Contextual Baselines

The main experiments use $B = 500,000$ training shots, giving approximately 4–5 shots per measurement under uniform allocation. We also perform a budget sweep over $B \in \{100\text{K}, 200\text{K}, 500\text{K}, 1\text{M}, 2\text{M}, 5\text{M}\}$ on PathMNIST, averaged over 30 independent noise realizations; the one-shot floor in Eq. (4) requires $B \geq NM = 90,000$, so smaller budgets are infeasible and not evaluated. Two baselines anchor the accuracy scale: a classical RBF SVM on the 6-dimensional PCA features and exact PQK using noiseless Pauli expectations, both using the same cross-validation and α grid as the finite-shot methods.

G. Evaluation Protocol

Each allocation strategy is evaluated over 30 independent runs. Training and test indices, exact PQR features, kernel structure, ridge parameter α , and classifier C are fixed across all methods on the same dataset. Only the measurement-noise realization changes across runs. The classifier parameter C and ridge parameter α are selected by 3-fold stratified cross-validation using the exact kernel, which avoids confounding allocation effects with hyperparameter changes. Adaptive allocation is applied only to training features, keeping the noise level on the test kernel identical across methods. The RBF bandwidth is fixed at $\gamma = 0.5$, which avoids introducing another source of variation across allocation methods, although it also means the reported accuracies are not bandwidth-optimized [6].

a) Statistical reporting. We report mean $\pm$ std. accuracy, paired two-sided t -tests, 95% CIs for the mean paired difference, and paired Cohen’s d_z . The primary BNKE-S vs. BNKE-U comparison uses Holm step-down correction across the five datasets. These are conditional on the fixed split, so they reflect robustness to measurement noise, not sampling variability.

Main results at $B = 500,000$ shots. Table I reports mean classification accuracy $\pm$ standard deviation over 30 paired runs for all five allocation, together with the two contextual baselines, at the main budget $B = 500,000$ training shots. The rightmost columns of Table I summarize the primary BNKE-S vs. uniform allocation on all five datasets.

b) Per-dataset results. BNKE-S achieves the highest finite-shot accuracy on PathMNIST (0.320), an improvement of $\Delta = +0.046$ over Uniform ($d_z = 2.96$), the largest effect observed across the five datasets; BNKE-SV performs comparably (0.317), and neither BNKE-V nor BNKE-B improves meaningfully over Uniform. On DermaMNIST, where PQR features are most competitive with the classical PCA baseline, BNKE-S reaches 0.630 ($\Delta = +0.013$, $d_z = 1.05$) and BNKE-SV is close behind at 0.627; BNKE-V and BNKE-B both fall 0.010 below uniform. PneumoniaMNIST gives BNKE-S its smallest effect (0.668, $\Delta = +0.015$, $d_z = 0.70$) and its widest confidence interval, though the gain still survives Holm correction ($p_H = 5.8 \times 10^{-4}$) and is confirmed by a Wilcoxon test ($p = 1.0 \times 10^{-3}$); this is also the one dataset where BNKE-SV edges out BNKE-S, reaching 0.671, and both baselines remain well above every finite-shot method here. The two held-out benchmarks tell the same story: OrganAMNIST reaches 0.322 ($\Delta = +0.022$, $d_z = 1.29$) and BloodMNIST reaches 0.420 ($\Delta = +0.019$, $d_z = 1.27$), so the effect is not limited to the three original datasets.

Sensitivity is the dominant allocation signal. BNKE-S improves over uniform on all five datasets, surviving Holm correction in every case (largest adjusted $p = 5.8 \times 10^{-4}$; Table I), with effect sizes from $d_z = 0.70$ on PneumoniaMNIST to 2.96 on PathMNIST. A Wilcoxon signed-rank test agrees with the t -test throughout, so the conclusion does not rest on normality. Note also what the winning weight omits: Eq. (7) uses feature geometry alone, no labels, classifier outputs, or margins. BNKE-SV differs from BNKE-S by at most 0.003 on any dataset, with no significant paired difference ($p \geq 0.21$);

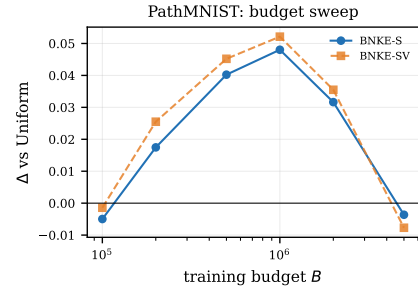


Fig. 3. Budget sweep on PathMNIST, 30 paired runs per point. Both variants improve over Uniform in the intermediate regime and are indistinguishable from it at the extremes.

we recommend BNKE-S as the simpler of two indistinguishable choices. Variance-only and boundary-only weighting give no such benefit: BNKE-V falls below uniform on DermaMNIST (-0.010), and BNKE-B falls below on DermaMNIST and PneumoniaMNIST. Non-uniform allocation is not beneficial in itself; the gain here is sensitivity-specific.

Figure 4 relates accuracy to surrogate loss at $B = 500,000$. Across all 150 per-run points, lower surrogate loss accompanies higher accuracy (Pearson $r = -0.86$, Spearman $\rho = -0.78$), reported descriptively since the points cluster by method and run. BNKE-S attains both the lowest mean surrogate loss and the highest mean accuracy among finite-shot methods.

Budget sweep. Figure 3 shows the effect of the total training budget on PathMNIST for BNKE-SV, BNKE-S, and Uniform. Each point is averaged over 30 noise realizations, matching the main experiment; 500K is reported only in Table I. Methods share a per-run noise seed, so intervals and tests derive from paired differences; these sweep tests are exploratory and not multiplicity-adjusted.

Three regimes emerge. At the lowest feasible budget (100K, about 1.1 shots per entry) BNKE-S is indistinguishable from uniform (-0.005 , 95% CI $[-0.012, +0.002]$); the weights do not degrade here, being independent of B , but the $\max(1, \cdot)$ floor binds on nearly every entry and forces the realized allocation close to uniform. Between 200K and 2M the paired intervals exclude zero at every point, rising from $+0.018$ to a peak of $+0.048$ near 1M before falling to $+0.032$: loose enough to depart from uniform, tight enough that placement matters. By 5M the advantage is gone, BNKE-S giving -0.004 (CI $[-0.015, +0.008]$), with BNKE-SV within 0.008. This is a vanishing benefit, not a reversal. Once uniform allocation estimates every entry well, sensitivity guidance adds nothing, consistent with Proposition 2: the surrogate L is nearly indifferent to error on low-weight entries, but classification is not. This crossover was measured on PathMNIST alone.

The same relationship holds across budgets: as B increases, surrogate loss falls and accuracy rises (Pearson $r = -0.86$, Spearman $\rho = -0.96$).

How much of the finite-shot penalty is recovered? Operating at 4–5 shots per entry carries a substantial cost: on PathMNIST, exact PQR reaches 0.612 against 0.274 uniform allocation, a penalty of 0.338. Sensitivity-guided allocation recovers 13.6% this penalty on PathMNIST, 27.1% on DermaMNIST, 12.1%

TABLE I

CLASSIFICATION ACCURACY (MEAN $\pm$ STD, 30 PAIRED RUNS) AT $B = 500,000$ TRAINING SHOTS ACROSS FIVE MEDMNIST DATASETS. **BOLD** = BEST FINITE-SHOT METHOD. CLS. (CLASSICAL RBF SVM) AND EXACT (NOISELESS PQQ) ARE CONTEXTUAL REFERENCES, DETERMINISTIC GIVEN THE FIXED SPLIT. Δ AND d_z COMPARE BNKE-S AGAINST BNKE-U (PAIRED, 30 RUNS); ALL FIVE DIFFERENCES ARE SIGNIFICANT AFTER HOLM CORRECTION ($p_H \leq 5.8 \times 10^{-4}$). CI IS THE 95% CONFIDENCE INTERVAL FOR Δ_{S-U} . A WILCOXON SIGNED-RANK TEST AGREES WITH THE t -TEST ON EVERY DATASET.

Dataset	α	BNKE-U	BNKE-V	BNKE-B	BNKE-S	BNKE-SV	Cls.	Exact	Δ_{S-U}	95% CI	d_z
PathMNIST	0.05	0.274 $\pm$ 0.013	0.276 $\pm$ 0.010	0.280 $\pm$ 0.012	0.320 $\pm$ 0.013	0.317 $\pm$ 0.011	0.636	0.612	+0.046	[.040, .052]	2.96
DermaMNIST	0.01	0.617 $\pm$ 0.008	0.607 $\pm$ 0.008	0.608 $\pm$ 0.008	0.630 $\pm$ 0.008	0.627 $\pm$ 0.011	0.489	0.665	+0.013	[.008, .018]	1.05
PneumoniaMNIST	0.05	0.653 $\pm$ 0.016	0.646 $\pm$ 0.015	0.642 $\pm$ 0.012	0.668 $\pm$ 0.013	0.671 $\pm$ 0.016	0.870	0.777	+0.015	[.007, .023]	0.70
OrganAMNIST	0.01	0.300 $\pm$ 0.012	0.294 $\pm$ 0.011	0.297 $\pm$ 0.011	0.322 $\pm$ 0.012	0.321 $\pm$ 0.013	0.609	0.543	+0.022	[.016, .029]	1.29
BloodMNIST	0.50	0.401 $\pm$ 0.012	0.395 $\pm$ 0.012	0.401 $\pm$ 0.011	0.420 $\pm$ 0.012	0.417 $\pm$ 0.014	0.677	0.677	+0.019	[.014, .025]	1.27

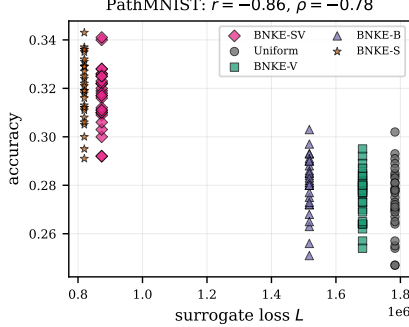


Fig. 4. Surrogate loss vs. classification accuracy across allocation methods at $B = 500,000$ shots, all 30 runs per method. Lower surrogate loss is associated with higher accuracy (Pearson $r = -0.86$, Spearman $\rho = -0.78$); the points are clustered by method and run, so no significance test is reported.

on PneumoniaMNIST, 9.1% on OrganAMNIST, and 6.9% on BloodMNIST—a useful but minority share, and not a substitute for a larger budget, despite the large effect sizes in Table I (d_z up to 2.96). **Robustness, conditioning, and the MSE accuracy gap.** To test robustness beyond a single split, we repeated BNKE-S vs. Uniform on five independently drawn train/test splits per dataset (10 noise realizations each, C and α re-tuned per split). BNKE-S leads uniform in all 25 split-dataset combinations, mean advantage +0.020 (+0.031 Path, +0.021 OrganA, +0.019 Blood, +0.015 Derma, +0.014 Pneumonia), with 20 of 25 individual comparisons reaching $p < 0.05$ (exceptions: three PneumoniaMNIST, two BloodMNIST splits). Given that absolute accuracy is itself strongly split dependent on uniform allocation on PathMNIST ranges from 0.276 to 0.412 a consistently positive difference is a notable finding.

Without ridge regularization, exact PQQ kernels appeared weaker than noisy finite-shot kernels on DermaMNIST (best exact PQQ 0.495 against 0.645 for uniform at two shots per measurement). The exact DermaMNIST kernel has condition number $\sim 2.1 \times 10^{18}$; with the CV-selected $\alpha = 0.01$ the exact result rises to 0.665 and the anomaly disappears, supporting ridge stabilization as a standard step in PQQ pipelines.

Finally, on PathMNIST, uniform allocation achieves the lowest raw kernel matrix MSE (0.424), whereas BNKE-SV (0.447) and BNKE-S (0.449) have higher MSE despite better classification accuracy—BNKE concentrates shots on kernel-sensitive entries and accepts larger errors elsewhere, so raw kernel MSE and downstream accuracy need not move together.

VI. DISCUSSION

A. Interpreting the Allocation Signals

a) *Why sensitivity outperforms variance, and why boundary proximity alone fails.*: Measurement variance peaks at $f_{ik} \approx 0$ but says nothing about whether that coordinate separates nearby samples in the RBF kernel. Sensitivity targets this directly: w_{ik}^{sens} grows large only when observable k meaningfully differentiates sample i from its neighbors, which is not what measurement variance tracks. BNKE-B fails for a related reason. Its weight u_i scales every observable of an uncertain sample equally, though only some actually separate it from its neighbors—a granularity mismatch, not a PCA/kernel inconsistency.

b) *Why the composite does not improve on sensitivity alone.*: BNKE-SV does not meaningfully improve on BNKE-S on any dataset (Sec. V), despite its variance term coming from exact, noise-free features. Variance likely just does not track kernel geometry: it up-weights coordinates that are hard to estimate but not decision-relevant, diluting the sensitivity signal. We prefer BNKE-S for simplicity, not demonstrated superiority; Proposition 3 is left unsupported by the data rather than contradicted.

B. Scope of the Quantum Contribution

This work does not claim that quantum kernel methods outperform classical alternatives on the datasets studied. On four of the five datasets, the classical RBF baseline computed on the same 6-dimensional PCA features exceeds every quantum variant tested here, in places by a wide margin (0.636 vs. 0.320 on PathMNIST). This deliberately small, untuned pipeline does not attempt to close that gap. DermaMNIST is the one dataset where the quantum pipeline is competitive, though it remains a small, constrained model. The contribution here is narrower: given a fixed quantum kernel pipeline and budget, allocation can be improved over Uniform when the sensitivity weights are known in advance.

C. Practical Implications

Four points follow. Prefer sensitivity to variance: sensitivity-only weighting matches the composite on all five datasets with one fewer factor and no labels, though as evaluated it remains an oracle benchmark and a deployable version must estimate w_{ik}^{sens} from a pilot pass. Apply ridge regularization

as standard practice, since ill-conditioning distorts exact vs. finite-shot comparisons. Expect gains only between 200K and 2M shots: below that the one-shot floor leaves little to allocate, above it every entry is already well estimated. Finally, do not treat raw kernel MSE as a proxy for accuracy, since the two need not move together under non-uniform allocation.

D. Limitations and Future Work

The principal limitation is that the sensitivity weight w_{ik}^{sens} is computed from exact PQK features obtained through state-vector simulation, which is unavailable on physical hardware or under a genuinely budget-constrained deployment. A hardware implementation would require a pilot measurement pass to estimate these quantities, so the results reported here should be read as an upper bound rather than a deployable method; designing such a pilot, and optimizing the split between pilot and allocation budget, is a natural next step. The variants also differ in the information available to them: BNKE-U requires no side information, BNKE-B uses classical margins, BNKE-V uses exact variances, and BNKE-S and BNKE-SV use sensitivity derived from the exact kernel. The ablation therefore reflects both allocation quality and the value of the information each variant has access to. The main comparison in Table I varies only the measurement-noise realization, holding the split, PCA, features, and hyperparameters fixed at seed 42; the split-robustness check in Sec. V extends this to five independent train/test draws, with BNKE-S ahead of uniform in all 25 cases. Only five splits and ten noise runs per split were examined, however, so generalization to other pipelines has not been established. It also remains an open question whether the dominance of sensitivity extends to deeper circuits, larger systems, other kernels, or non-medical data. Computing w_{ik}^{sens} carries a cost of $O(N^2M)$, tractable at the scale used here but a potential bottleneck as N grows; Nyström subsampling or blockwise approximation could offer a remedy at some cost in accuracy. A gap between theory and practice also remains, since the theory identifies BNKE-SV as the surrogate optimizer, while BNKE-S is the strongest empirical variant. Finally, all results rely on exact state-vector simulation with analytically injected shot noise, validated against exact Bernoulli sampling in Sec. IV-D. No hardware experiments were conducted, and circuit-level noise sources such as crosstalk, SPAM error, or drift are not included.

VII. CONCLUSION

We introduced BNKE, a weighted allocation framework for adaptive shot allocation in projected quantum kernel estimation, built on a Neyman-style allocation rule derived from a weighted surrogate loss together with a first-order kernel sensitivity argument. Across five MedMNIST datasets, kernel sensitivity proves to be the dominant allocation signal: BNKE-S significantly outperforms uniform allocation on all five datasets, while variance-only and boundary-only strategies provide no consistent gain. The allocation weight itself relies on no labels, classifier outputs, or margins. The advantage is largest in the shot-scarce regime and disappears once the per-entry shot count

grows large, recovering between 7% and 27% of the accuracy lost to finite sampling, a meaningful improvement though not one that closes the gap to the classical baseline studied here. A second finding is that kernel reconstruction fidelity and downstream accuracy can pull in different directions under non-uniform allocation: BNKE may leave a higher kernel MSE than Uniform while still classifying more accurately, since it allocates shots to the coordinates with the greatest first-order influence on the kernel rather than minimizing raw reconstruction error. Ridge regularization is also shown to be necessary for stable PQK evaluation, as the DermaMNIST anomaly demonstrates. This paper makes no claim of quantum advantage: all experiments are simulated, and the contribution is one of measurement efficiency within a fixed PQK pipeline. The weights driving the strongest variant are oracle quantities, so estimating sensitivity under a genuinely finite budget remains the principal obstacle to deployment. These results establish that kernel sensitivity carries enough allocation-relevant information to make that estimation problem worth pursuing; larger qubit counts, deeper circuits, and surrogate objectives more closely tied to classification loss are natural directions for future work.

ACKNOWLEDGMENT

We thank the anonymous reviewers for their feedback. This research was in part supported by the US NSF under Grants: SHF-2210744, CSR-2549533, and AMPS-2229073.

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